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Research paper

Employing convolutional neural networks and explainable artificial intelligence for the detection of seizures from electroencephalogram signal

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ABSTRACT

A seizure is a rapid, uncontrolled burst of electrical activity in the brain. Epilepsy is a neurological disorder caused by repeated seizures. Effective management requires early detection. This research aims to create a convolutional neural network (CNN) and explainable artificial intelligence (XAI) integrated system for epileptic seizure detection, using techniques such as Shapley additive explanation (SHAP) for better interpretability. The relevant data is acquired from a variety of electroencephalogram (EEG) dataset to facilitate the development of the model. To identify the EEG data, the proposed system combines deep learning models with feature extraction technique. Model visualization and XAI approaches like feature importance analysis from SHAP values offer clear insights into the model's decision-making process. Evaluation criteria like specificity and accuracy are used to assess the models' performance. This framework's objective is to create simple seizure detection systems that assist early epilepsy patient identification and individualized treatment plans. This research aims in opening the door to improved patient care and treatment by developing the field of epilepsy seizure detection.

1. Introduction

Seizures are a serious medical disease that has to be detected and treated swiftly. They are defined by aberrant electrical signals in the brain during the first month of life. Instance rates are greater in premature infants and those with neurological disorders. They occur in around one to five out of every 1000 births. In order to avoid permanent brain damage and reduce long-term neurological consequences, it is imperative that seizures be identified and treated immediately [1]. Despite medical advancements, diagnosing neonatal seizures is challenging due to their subtle nature. Symptoms vary widely, from subtle movements to overt motor activity, often resembling other non-specific conditions like feeding difficulties and gastrointestinal disturbances. Hence, diagnosis of neonatal seizures heavily depends on subjective interpretation of clinical signs and electroencephalogram (EEG) recordings by trained clinicians [2].

Neonates' brain activity can now be mainly observed via EEG, which provides significant information into the dynamics of neural function [3]. Electrographic seizures can be identified with continuous EEG monitoring regardless of whether they always have obvious clinical symptoms. Traditional techniques for detecting seizures, however, mostly rely on the visual interpretation of EEG data, which can be labor-intensive, time-consuming, and inconsistent. Furthermore, visual EEG analysis has poor accuracy and precision for seizure identification, especially in situations of mild or transient seizures [4]. With Machine Learning (ML) and Artificial Intelligence (AI) advancements, interest grows in enhancing EEG-based seizure detection. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) show promise due to their success with sequential data [5]. CNNs excel at capturing spatial dependencies in multi-channel EEG data, while RNNs model temporal dynamics in seizure patterns. Researchers aim to leverage CNNs and Recurrent Neural Networks (RNN) complementary strengths to develop more robust accurate models for neonatal seizure detection.

In addition to achieving high predictive accuracy, ensuring the transparency and interpretability of machine learning models is crucial, particularly in healthcare applications where decisions directly impact patient outcomes [6]. To address this challenge, researchers have begun to explore Explainable Artificial Intelligence (XAI) techniques, which aim to provide insights into the inner workings of black-box models and generate interpretable explanations for model predictions. By integrating XAI techniques into CNN and RNN architectures, researchers hope to enhance the interpretability of model decisions and facilitate clinical adoption and trust in automated seizure detection systems.

The creation of precise and trustworthy seizure detection models is

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crucial for the treatment of newborns. To minimize long-term neurological problems and avoid permanent brain damage, seizures must be identified early and treated promptly. Diagnosing seizures in infants can be particularly difficult since they frequently appear as fleeting motor activity or modest motions that are readily confused with other non-specific illnesses. As a result, the dependence on the subjective clinical interpretation of EEG recordings and symptoms highlights the critical need for more effective and reliable techniques.

A vital component of evaluating brain activity is EEG monitoring, which provides priceless insights into the underlying neuronal dynamics. On the other hand, traditional seizure detection techniques rely heavily on the human visual interpretation of EEG data, which has its own set of drawbacks concerning accuracy, reliability, and versatility. The need for reliable and automated seizure detection methods is highlighted by the labor-intensive nature of visual EEG analysis and its vulnerability and insensitivity towards human error. The current convergence of ML and AI approaches has demonstrated potential to transform seizure detection strategies. CNNs have become highly effective tools because to their capacity to identify time and spatial relationships in EEG data, respectively. Researchers have aimed to create complex models that minimize false positives and negatives while effectively identifying seizures by utilizing the complimentary capabilities of CNN.

Nevertheless, it is essential that good prediction performance, accountability, and interpretability are obtained when using AI models in therapeutic contexts. This problem can be solved by using XAI tools, which provide physicians, insight into predictions made by models and help them comprehend the underlying processes that influence decision-making. Scholars aim to improve the interpretability and reliability of automated seizure detection systems by integrating XAI techniques into CNN and RNN designs. This would enable the systems to be easily incorporated and used regularly in healthcare settings. The success of these models depends not only on their predicted accuracy but also on their capacity to give doctors useful insights by exploring the field of AI-driven seizure detection. By using XAI methodologies, healthcare personnel have the ability to comprehend how AI models make judgments, which boosts their trust in the diagnostic process. Additionally, XAI helps identify clinically useful diagnostics by highlighting the key characteristics and patterns that underlie seizure occurrences. This has the potential to completely transform the diagnosis and treatment of epilepsy.

Predictive models are useful in the field of AI-driven seizure diagnosis for reasons more than just accuracy. Their interpretability and the insights they provide to healthcare professionals are important factors. Practitioners can now see into the AI models' decision-making methods with more clarity thanks to XAI approaches, which increases their confidence in the accuracy of the diagnostic results. Because of this openness, medical staff members are able to test and improve diagnostic choices by understanding the rationale behind model predictions and relying on both medical expertise and AI-driven insights. Furthermore, the incorporation of XAI methodologies enables medical professionals to extract clinically significant insights from the intricate EEG data that is available.

Current methods for identifying seizures mostly rely on an EEG monitor in conjunction with clinical observation. Physicians keep a close eye to spot any unusual movements, behavioral shifts, or other symptoms that might indicate seizure activity. Simultaneous real-time data on neural activity is provided by constant EEG monitoring, which enables doctors to diagnose electrographic seizures even in the absence of obvious clinical symptoms. These conventional methods, however, are time-consuming, subjective, and vulnerable to variation across observers in practical use cases.

Computerized seizure detection methods have been developed to help doctors identify seizures more reliably and effectively in addition to clinical examination and EEG monitoring. In order to evaluate EEG data and identify seizure patterns, these systems usually make use of machine learning methods, such as CNNs and RNNs. Although seizure detection using these AI-driven methods appears to be improving, issues with their dependability, clarity, and smooth integration into clinical workflows still need to be resolved. As a result, there is an increasing need to create trustworthy and transparent AI models that may improve healthcare providers' diagnostic skills without compromising patient safety or quality of therapy.

In this study, XAI methods are combined with CNN to provide a distinctive approach for seizure identification. It is aimed that this combination of powerful deep learning models and straightforward interpretability techniques might improve seizure detection models' comprehensibility and accuracy while also promoting acceptance and confidence in clinical contexts. Our objective is to close the gap between state-of-the-art machine learning technology and the practical requirements of clinical practice by utilizing CNNs to recognize complex spatial relationships in EEG data and XAI approaches to clarify the foundational decision-making processes.

The paper is structured to cover various key aspects of the proposed system, including data preprocessing methods, feature extraction, the architecture, training and evaluation of CNN model. The integration of XAI methodologies for model interpretation, and an in-depth analysis of the results have been obtained and displayed in the results sections. Furthermore, assessments of the findings' significance and possible future study avenues have been included. This organized approach allows for a systematic understanding of the methods employed, the insights gained, and the significance of the study's contributions to the field of epileptic seizure detection and interpretation.

2. Literature survey

The existing system explored in this paper focuses on classifying EEG signals for epilepsy detection. It uses fuzzy classifiers with fuzzy decision trees and random forests. Feature extraction, pattern classification, and dimensionality reduction are employed, along with the use of Principal Component Analysis (PCA). The study achieves an accuracy of 99.3 % in classifying EEG signals. However, it has limitations of signal preprocessing exploration, and the choice of evaluation metrics, and does not specify further explorations [7].

One other study explores methods to enhance epileptic seizure classification through EEG signals. Support Vector Machine (SVM), K-Nearest Neighbours (KNN), Naive Bayes is used in this paper. The technique decomposes multichannel EEG signals into Intrinsic Mode Functions (IMF) using empirical mode decomposition (EMD), followed by IMF selection and feature extraction. Using EMD, accuracies are from 94.56 % to 96.8 %. Limitations include the need for validation on larger datasets, sensitivity to a variety of patients [8].

The study aims to enhance EEG signal classification accuracy, specifically targeting epileptic seizure detection. It makes use of SVM, KNN classifiers, and Multilayer Perceptron (MLP). The basis for feature extraction, which includes frequency, surfaces, K-means clustering with Local Ternary Pattern (LTP) adaptation, and maximum peaks, is a Spectrogram produced by using the Fourier Transform (STFT) on the signals. The study emphasizes the need for multi-class classification to distinguish distinct types of epilepsy, achieving a respectable accuracy rate of 100 % in some situations with fewer features than prior works. Even with its effectiveness, it has drawbacks, such as just taking into account the biggest peak from spectrogram columns, which means more research into nearby peaks is required for a thorough analysis [9].

The study delves into predicting epileptic seizures, particularly targeting the pre-ictal state, through EEG signals. CNN is used to compute the achievements of neural network approaches for this forecasting job. To obtain relevant features from EEG signals—which include both scalp and intracranial data for improved prediction accuracy—the research used Fourier transform and EMD techniques, focusing on CNN architecture. The study finds areas for improvement, including improving preprocessing for a higher signal-to-noise ratio, reducing the number of parameters, and addressing the patient-specific nature of the prediction method [10].

Another research focuses on real-time diagnosis of epilepsy utilizing a Field Programmable Gate Array (FPGA)-based technique for classifying seizure types. Using 822 brain waves from the dataset it trains, verifies, and tests the system. Using time and frequency analysis, the system extracts five primary characteristics from EEG data that are then subjected to statistical analysis. With a precision of 95.14 %, real-time seizure categorization is made possible on the FPGA by implementing a 5–12–3 MLP artificial neural network (ANN). The study emphasizes how important it is to create mobile and scalable expert illness categorization systems for improved epileptic seizure diagnosis [11].

The research aims to automatically classify abnormal EEG signals from normal ones in detection seizures with minimum computational costs. The procedure uses a 54-Discrete Wavelet Transform (DWT) mother wavelet analysis of EEG signals and uses SVM, KNN, ANN, and Naive Bayes classifiers in conjunction with genetic algorithms for feature selection. By utilizing the designated classifiers for examining 14 distinct combinations of two-class epilepsy detection, the genetic algorithm helps reliably identify and categorize epileptic episodes. The study shows the significance of feature selection and extraction stages, which have significant effects on the system's sensitivity, with average accuracies ranging from 97.15 % to 97.82 % [12].

Another research focuses on defining the correct parameters. Employing SVM and ANN classifiers, the methodology involves using a 5-level Db4 DWT and statistical calculations for extracting relevant parameters, with genetic algorithms aiding in feature selection. Achieving remarkable accuracies of 100 % for both SVM and ANN in the 2-class classification, and 98.7 % for SVM and Artificial Neural Network (ANN) in the 3-class classification, the study acknowledges limitations due to the small size of the dataset utilized [13].

One paper on the topic, addresses the crucial need for accurate seizure detection in epilepsy management. Raw EEG data complexity demands advanced processing; DWT extracts relevant information by dividing the EEG signal into five sub-bands. Genetic algorithm optimization enhances feature selection, boosting discriminative power. Seizure detection employs ANN and SVM classification techniques. Comparative analysis demonstrates superior accuracy of the proposed strategy in epileptic seizure detection. By integrating cutting-edge signal processing and machine learning, the study enhances diagnostic precision, promising improved patient outcomes in epilepsy management [14].

The study focuses on dynamic epileptic seizure detection in long-term EEG. EEG data undergoes preprocessing using SVM classifier to eliminate noise and artifacts, followed by segmentation of each channel at a specified non-overlapping window length. Successive decomposition index (SDI) computation achieves 97.53 % sensitivity with SVM classification. Comparative analysis favors SDI for its superior computational efficiency, robustness, and wider applicability. However, practical implementation on a hardware platform is necessary for real-time seizure detection. Improvements in multichannel EEG feature extraction real-time processing are recommended to enhance computational speed and algorithm performance [15].

The study focuses on classifying seizure and non-seizure events. The Tunable Q-wavelet transform (TQWT) breaks down the data signal into smaller regions using SVM and Random Forest (RF) classifiers, making it easier to extract time-frequency-based information including temporal measurements and entropy. SVM and RF are used to handle the resulting dataset; RF performs better than SVM, reaching 91.5 % sensitivity and 93 % accuracy. The paper notes limits such as prohibited dataset representation, lack of external validation on independent datasets, and generalizability to other datasets, providing opportunities for more research and improvement even if the results show promise for medicinal use [16].

The research aims to identify pre-seizure and seizure states in EEG signals for accurate prediction. The study combines frequency domain characteristics with Power Spectral Density, time domain features using a fuzzy classifier and pattern-adapted wavelet transform. The massive feature collection is managed using feature selection, and wavelet, frequency, and time domain data are combined in a fuzzy logic model to accomplish classification. EEG signals are classified into normal, pre-seizure, and seizure states using fuzzy rules and triangular membership functions. The rating accuracy is 96.02 % for pre-seizure state detection and 96.48 % for seizure identification. Nonetheless, the research proposes further ways to enhance classification accuracy, namely the use of sophisticated techniques like Fuzzy Inference Systems (FIS), accurate feature selection, sophisticated and appropriate classifiers, and efficient non-specific algorithms [17].

The study focuses on di normal EEG signals from seizures in ictal and inter-ictal stages. The study uses a Computer-Aided Diagnostic (CAD) approach combining mathematical coding and DWT to efficiently identify epileptic seizures in EEG signals. It does this by utilizing both linear and non-linear ML algorithms, such as KNN and MLP. The process includes breaking down the EEG signals using DWT and then eliminating non-significant coefficients according to predetermined thresholds. Then, using arithmetic coding, significant wavelet coefficients are converted into bit streams to provide a feature set that is fixed for ML classifiers. The paper proposes to address possible computational costs and improve classification accuracy further by utilizing bigger clinical databases and including feature selection procedures such as PCA following extraction, achieving an overall accuracy of 100 % [18].

The study addresses data labeling in training EEG-based seizure detection algorithms. The modules for supervised learning (SL) and Unsupervised Learning (UL) are integrated. For preliminary seizure screening, the UL module uses silhouette coefficient-based anomaly detection, adaptive segmentation, isolation forest-based anomaly detection, and Amplitude-integrated EEG (aEEG) extraction. This module minimizes the labor associated with data labeling by rapidly identifying seizure, non-seizure regions and possible seizure regions. The SL module improves seizure detection for undetermined people by using Easy Ensemble, which may lower generalization error. 92.62 % mean accuracy, 95.55 % mean sensitivity, and 92.57 % mean specificity have been determined by the study. The CHB-MIT dataset has obstacles, such as missing information on certain seizure types and the need for improved internal cluster validity indices [19].

The study aims to efficiently identify epileptic seizures for early-stage detection and treatment. Using random search cross-validation, nine classifiers—including SVM, RF, and MLP—are used to create and assess models both with and without hyper parameter adjustment. Numerous effectiveness parameters, including accuracy, precision, recall, specificity, F1-Score, and receiver operating characteristic curve (ROC), are taken into account in comparative comparisons. With a high accuracy of 97.86 % for SVM, 97.50 % for RF, and 97.26 % for MLP, the research indicates that deep learning may be used to create systems that are more resilient [20].

The study focuses on developing a group of classifiers that are bio-inspired heuristic-tuned for epileptic seizure detection. Scalable hypothesis tests are used to extract features from EEG signals using SVM, RF, and KNN classifiers. The most significant features are chosen using RF. For seizure detection, an ensemble of three classifiers adjusted with the Crow Search Algorithm (CSA) is utilized. The research claims an accuracy rate of 83 % overall. For improved performance, it does, however, recommend using better feature selection techniques and taking into account a larger number of classifiers in the ensemble [21].

The study focuses on leveraging machine learning and deep learning algorithms to enhance the automatic identification of epileptic seizures. (SVM, CNN), (MLP, CNN), and (SVM, MLP, CNN) are the three hybrid ML classification networks that are examined. Following data processing, the model extracts feature of time domain by using EMD. Effective seizure prediction is achieved by the model by the use of a hybrid machine learning strategy that combines CNN with SVM and MLP. With wavelet transformation and sample entropy, the model uses posttraining to precisely recognize the ictal and pre-ictal regions of seizures. An accuracy score of 89 % places the hybrid CNN, SVM, and MLP algorithm above other hybrid algorithms. Nevertheless, the study recommends developing strategies for precisely determining the firing site of the seizure as well as enhancing the model's ability to detect confined dispersed partial seizures [22].

The paper aims to distinguish between normal seizures, partial seizures, and generalized seizures based on EEG data. It makes use of the multiple ML algorithms. All of the methods generate and compare common performance indicators. The computational processes entail feature extraction, classification, and data preparation, culminating in a thorough assessment of seizure classification effectiveness [23]

One such study introduces an ANN framework for seizure detection with optimized performance. Three primary processes comprise the methodology: a two-layer approach for feature extraction; hyper parameter optimization via Bayesian optimization; and seizure classification via an artificial neural network with binary classification. Four paediatric patients' EEG records are taken into consideration, and Bayesian optimization is used to fine-tune the ANN design and parameters. Input data are scaled using standard scalar methods, and binary cross-entropy loss is employed in the classification process. 99.50 %, 99.32 %, and 99.72 %, respectively, are the greatest levels of accuracy, sensitivity, and specificity that have been attained [24].

The research presents a generalized approach for classifying seizure types in epileptic patients using a CNN. Pre-processing is performed consisting of 3050 seizures from 8 classes. 1-second signals are segmented using the Fourier transform and frequency filtered. Training a Residual Network (ResNet)-based model with an attention module from scratch emphasizes the significance of features. Through three-fold patient-wise cross-validation, the model is assessed while taking signal-level and segment classifications into account. Training is done using the Adam optimizer and weighted cross-entropy loss to correct class imbalance. At the signal level for different types of seizures, the CNN results get a F1 score of 69.33 %. The model's effectiveness in categorizing tonic-clonic seizures, generalized non-specific seizures, and complicated partial seizures, however, points to more room for development [25].

The study focuses on detecting seizures accurately and quickly in order to differentiate between pre-ictal, inter-ictal, ictal, and healthy states. Using native real-time EEG data capture, signal segmentation, and the DWT are all part of the process. Aspects including average, standard deviation, and the first components of the PCA are retrieved. A quadratic discriminant classifier is used for classification. For effective seizure detection, this method combines machine learning, feature extraction, and signal processing. On a dataset containing 12,500 segments, the method obtains a 99 % real-time epileptic seizure identification rate. The paper addresses the challenge of handling large-volume and high-dimensional EEG data for optimal computational performance. Long Short-Term Memory (LSTM), a deep learning model, and SVM, RF, and Decision Tree (DT) are used [26].

According to the study, essential characteristics (maximum, minimum, standard deviation, mean, median, and energy) are extracted using the DWT, and parameters are selected using variable analysis with threshold variations. The SVM approach performs best in order to gain higher precision in predictions. The study suggests incorporating more parameters [27]

The study intends to create a wearable device-based seizure detection that is both dependable and effective. A SVM classifier is used in this method. Pre-processing of the data, feature extraction from the time, frequency, or time-frequency (Wavelet) domains, binary classification using SVM, issue metrics evaluation, and model optimization are the four stages of a suggested seizure detection system. The technique attains an average accuracy of 97 % and sensitivity of 100 % [28].

The detailed study addresses the challenges of manual detection and classification by radiologists, advocating for automated methods due to previous limitations in computational time and accuracy. The Adaptive Sine Cosine Algorithm - Whale Optimization Algorithm - Extreme Learning Machine (ASCA-WOA-ELM) model is used in this suggested method. In terms of accuracy, specificity, sensitivity, and computing time, this combination of approaches beats other Extreme Learning Machine (ELM) models by maximizing ELM weights. It is noticed that a bigger data set cannot be used [29].

An intensive research paper speaks on precisely detecting seizures and classifying them into four levels: pre-ictal, ictal, inter-ictal, and normal. The XGBoost algorithm and a high-input resistance amplifier are used in the proposed seizure detection system to handle real-time data from an EEG sensor. When classifying signals into seizure and healthy states, XGBoost performs accurately because of its speed, regularization, feature importance assessment, and management of missing data. Raspberry Pi Pico is the hardware used in the implementation. 96 % total accuracy was attained with the following accuracy for each stage of the seizure: 88 % for the ictal stage, 84 % for the pre-ictal stage, and 91 % for the inter-ictal period [30].

Beyond classification accuracy, research in EEG-based seizure detection has to take into account other metrics in order to give a more thorough knowledge of model performance. In addition, there is insufficient attention paid to signal pre-processing techniques, which are essential for enhancing classification dependability and accuracy. Widespread application depends on the model's capacity to generalize across various patient groups and datasets. Furthermore, in order to build clinician trust, efforts must be made to improve model interpretability using XAI approaches. XAI methods have not been distinctively used to improve the model's understanding and this provides an opportunity to enhance the existing study to achieve more in the given domain.

This paper, aims to overcome these drawbacks by proposing a methodology that integrates CNN model with XAI to get higher accuracies while understanding and validating the developed model using the XAI techniques. This approach increases the trust in the model and its comprehensibility thereby playing an important role in its clinical implementation.

3. Proposed methodology

The proposed methodology for seizure detection from EEG data using CNNs, and XAI techniques involves several interconnected steps aimed at optimizing model performance and interpretability.

(i) Data Preprocessing:

The dataset is segregated into 5 classes as per the signal collection obtained from the electrodes in the electroencephalogram process. EEG signals undergo preprocessing to enhance signal quality and extract relevant features. This includes de-noising techniques, artifact removal, and segmentation into smaller time windows. Feature extraction methods such as wavelet transforms and time-frequency analysis are to be applied to capture discriminative features.

(ii) Architecture Design:

The architecture is described elaborately in Fig. 1 and it comprises of CNN and XAI components. The CNN component captures spatial dependencies within EEG signals through convolutional layers, while the XAI component interprets the model and analyzes its decision making processes.

(iii) Feature Extraction:

To identify epileptic activity in the EEG signals, the focus shifts to specific frequency ranges that are well-established for EEG signals termed as delta, theta, alpha, beta and gamma with each having specified ranges and significance. Fourier transform analysis is done thereby

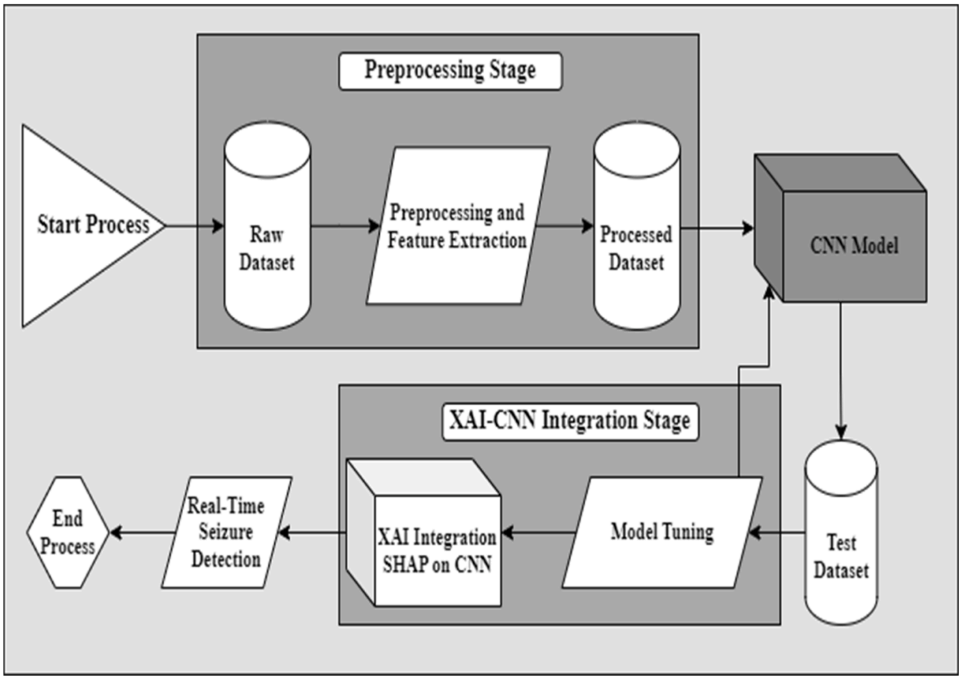

Fig. 1. Architecture Diagram of Seizure Detection System.

obtaining the five medically ascertained EEG bands – delta, theta, alpha, beta, gamma frequencies. To obtain the amplitude of EEG Signals in these ranges, power spectral density is integrated over the desired frequency range. This helps in observing epileptiform discharges in the frequency bands to confirm presence of epileptic seizures.

(iv) Model Training:

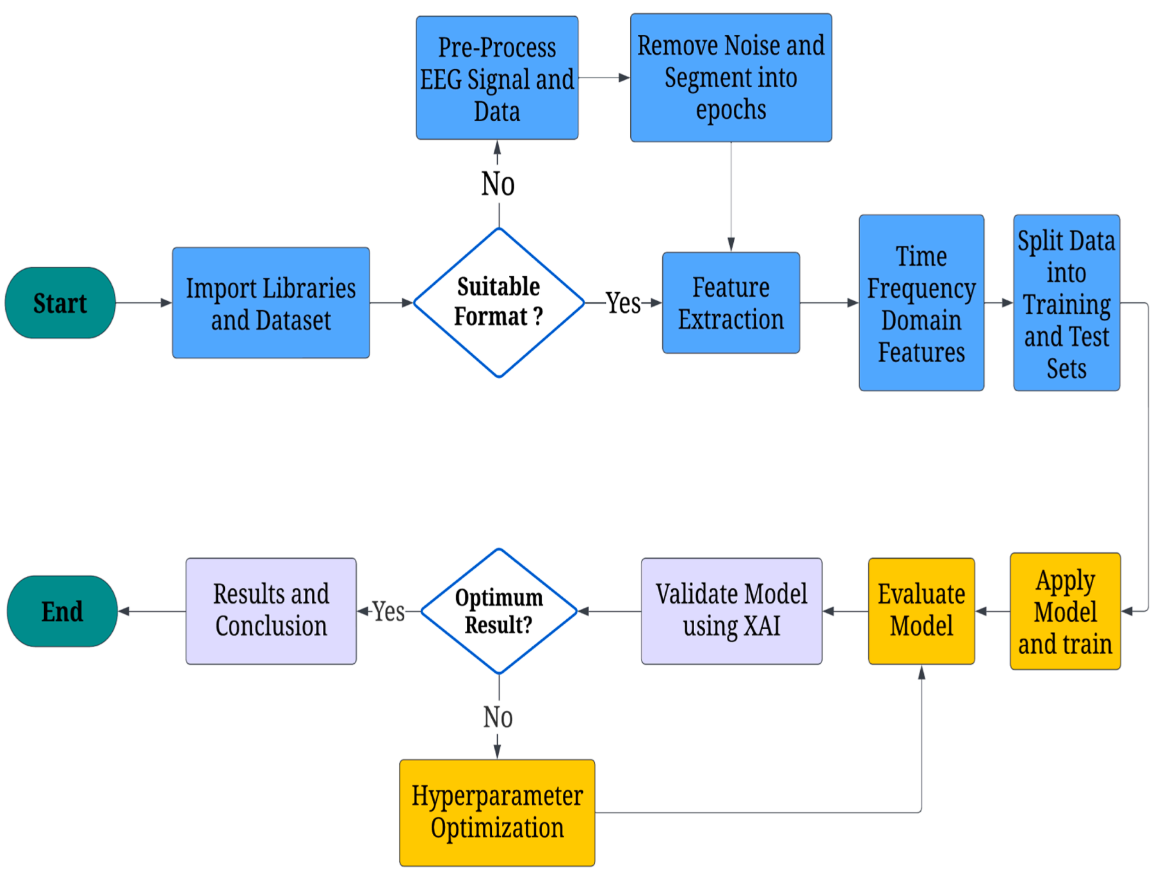

Fig. 2. Workflow of detecting epileptic seizures using XAI and CNN.

Preprocessed EEG data is segregated into training, validation, and test sets. The CNN architecture is then trained on the training data using appropriate loss functions and optimization algorithms. Hyper-parameter tuning techniques would be employed to optimize model performance. The model is trained for 10 epochs and then applied on a test set to get the binary classification of seizure or non-seizure.

(v) XAI Integration:

XAI techniques are used to interpret the models' predictions and provide insights into their decision-making processes. Methods like SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanation (LIME), or Layer-wise Relevance Propagation (LRP) are applied to attribute the model's decisions to input features and visualize their contributions.

(vi) Evaluation and Interpretation:

The trained models are evaluated on the test set using metrics like accuracy, sensitivity, specificity, and area under the curve (AUC)-ROC. Interpretability metrics, such as feature importance scores or saliency maps generated by XAI techniques, are used to assess the models' interpretability. This interpretation gives an analysis on how the data-points collected from the electrode impact the results of the model's predictions.

These 5 modules form the proposed system of the seizure detection system. Fig 2 shows a thorough workflow for detecting epileptic seizures using a deep learning architecture that combines EEG data with Explainable Artificial Intelligence (XAI) and Convolutional Neural Networks (CNN). Importing the required libraries and datasets is the first stage in the procedure. Next, pre-processing processes are performed to ensure that EEG signals are formatted appropriately. In order to separate pertinent signal characteristics, pre-processing techniques such as noise reduction and epoch segmentation are used if the data is not in the proper format. In order to get important data, including time-frequency domain features, which are inputs to the deep learning model, feature extraction is then carried out.

The data is then divided into training and testing sets, with the training set being used to train the model. Using the test data, performance is evaluated as part of the model evaluation process. Iteratively, hyperparameter optimization approaches are used to improve the model's performance if the outcomes are suboptimal. Following the acquisition of good findings, XAI approaches are used to validate the model. These techniques give the model's predictions interpretability, guaranteeing that the results are not only accurate but also comprehensible. Deriving outcomes and conclusions at the end of the workflow can provide light on the effectiveness of seizure detection and the interpretability of the model.

(a) Data Pre-processing:

The dataset used in this study consists of 500 recordings pertaining to the identification of epileptic seizures. Each recording is unique to each person and includes comprehensive details regarding their brain activity. To guarantee thorough coverage, EEG signals are recorded for 23.6 s throughout the data collecting process and then sampled into 4097 data points. Each recording is then divided into 23 chunks, each having 178 data points that correspond to 1-second intervals, to enable efficient processing.

After this thorough preparation step, the dataset is arranged into an 11,500 sample structured collection. Every sample contains EEG data that is captured at 1-second intervals and is labelled with whether a seizure or non-seizure event occurred. This methodical technique makes supervised learning easier and allows machine learning models to distinguish between seizure and non-seizure cases with accuracy.

The dataset is tested for missing values to avoid errors. Additionally, the dataset uses a balanced classification strategy to ensure that data on seizures and non-seizures are equally represented. It is imperative to have this balanced distribution in order to train and evaluate machine learning algorithms efficiently. The structured samples extracted from the original EEG recordings greatly improve the dataset's machine learning algorithm compatibility, increasing its usefulness for testing and training.

(i) Filtering:

Bandpass filters are used to effectively remove noise while maintaining the integrity of the EEG signal by selectively attenuating frequency components outside of the target range. Bandpass filtering improves the signal-to-noise ratio by focusing on certain frequency bands, such as alpha, beta, theta, and gamma waves, that are linked to brain activity. This procedure reduces artifacts and unnecessary interference while aiding in the isolation of pertinent brain activity patterns. By ensuring that only frequencies pertinent to EEG analysis are kept, bandpass filters help to improve the accuracy of seizure detection algorithms and facilitate further processing stages.

(ii) Segmentation:

In order to segment continuous EEG signals, they must be divided into smaller, discrete segments, or epochs, usually with a constant duration. This procedure makes it possible to analyse particular temporal intervals with precision, which makes it easier to identify ephemeral events like epileptic seizures. Researchers can more effectively focus on localized changes in brain activity and identify patterns associated to seizures by segmenting EEG data into manageable bits. Additionally, segmentation enables the use of particular algorithms and feature extraction methods for each epoch, improving the specificity and sensitivity of seizure detection systems. Therefore, segmentation is an essential preprocessing step that allows for more in-depth analysis and raises the precision of algorithms for EEG-based seizure identification.

(iii) Normalization:

In order to standardize the data's amplitude range, normalization is a vital preprocessing method in the interpretation of EEG signals. Normalization, as shown in Eqs. (1) and (2), involves scaling EEG signals to a similar range, like between 0 and 1 or −1 and 1, guarantees consistency between recordings and improves model convergence during training. By removing any biases caused by changes in signal strength, this procedure improves the data's suitability for comparison and interpretation. By avoiding features with enormous numerical values from controlling the learning process, normalization also helps reduce the impact of outliers and enhances the stability of machine learning systems.

In order to improve consistency and reliability of EEG data processing for seizure detection applications, normalization and scaling is given in Eq. (1)

Normalization

$$\widehat{x} = \frac{x - \mu}{\sigma} \tag{1}$$

Scaling

$$x_{scaled} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{2}$$

Where, $x \rightarrow$ represents the value of a single datapoint iterated over all the datapoints in the dataset

$\mu \rightarrow$ is the computed average of the datapoints

$\sigma \rightarrow$ is the computed standard deviation of the datapoints

(b) Feature Extraction
(i) Time-Domain Features:

These metrics provide insights into the fundamental properties of the signal by quantifying its variability, shape, and distribution over time. For example, kurtosis evaluates the signal's peak-ness or tailed-ness, whereas skew-ness quantifies asymmetry. Eqs. (3), (4), (5) and (6) calculate the time-domain features such as mean, standard deviation, skewness and kurtosis respectively. By examining these characteristics, it is possible to recognize unique patterns that point to epileptic seizures, which helps with diagnosis and treatment formulation. Furthermore, time-domain properties offer machine learning algorithms useful input, improving their capacity to reliably identify EEG data and automate seizure detection procedures.

Mean

$$\mu = \frac{\sum_{i=1}^{n} x}{n} \tag{3}$$

Standard Deviation

$$\sigma = \frac{\sqrt{\sum_{i=1}^{n} (x_i - \mu)^2}}{n} \tag{4}$$

Skewness

$$\gamma_1 = \frac{\sum_{i=1}^{n} (x_i - \mu)^3}{n\sigma^3} \tag{5}$$

Kurtosis

$$\gamma_2 = \frac{\sum_{i=1}^{n} (x_i - \mu)^4}{n\sigma^4} \tag{6}$$

Where, $x \rightarrow$ represents the value of a single datapoint iterated over all the datapoints in the dataset

$\mu \rightarrow$ is the computed average of the datapoints
$\sigma \rightarrow$ is the computed standard deviation of the datapoints
$n \rightarrow$ is the total number of the datapoints

(ii) Frequency-domain features:

These features entail breaking down signals into their frequency components using mathematical transformations such as the Fourier or Wavelet transforms. Features that can be gleaned from these decompositions include power spectral density, spectral entropy, and the energy contained in particular frequency bands. These characteristics help characterize EEG patterns linked to epileptic seizures and enable more precise seizure detection algorithms by offering insights into the distribution of signal power across various frequency ranges. Eq. (7) shows the calculation of Fourier Transform which helps in decomposing time-domain signals to frequency domain.

Fourier Transform (Continuous)

$$F(\omega) = \int_{-\infty}^{\infty} f(t).e^{-j\omega t} \tag{7}$$

Where

$F(\omega) \rightarrow$ represents frequency domain representation of the signal
$f(t) \rightarrow$ Original signal in the time domain
$e^{-j\omega t} \rightarrow$ Complex exponential function
$\omega \rightarrow$ Angular frequency

(iii) Time-frequency domain features:

It incorporates methods such as the continuous wavelet transforms (CWT) and STFT. These techniques allow for the simultaneous analysis of frequency and time information in EEG signals, which makes it possible to record brief variations in frequency content across time. Through the utilization of STFT or CWT, investigators can derive characteristics that mirror the dynamic evolution of the frequency composition of EEG data, offering significant understanding into the temporal dynamics of brain activity during epileptic episodes. Wavelet transforms can be calculated, as shown in Eq. (8), to categorize the signal in the frequency sub-bands.

Wavelet Transform (Continuous)

$$W(a,b) = \int_{-\infty}^{\infty} f(t).\psi^{*}\left(\frac{t-b}{a}\right) dt \tag{8}$$

Where,

$W(a,b) \rightarrow$ represents wavelet coefficients
$f(t)$ is the time domain original signal.
$\Psi \rightarrow$ complex conjugate of the wavelet function.
a and b are the scale and translation parameters

To identify epileptic activity in the EEG signals, the focus shifts to specific frequency ranges that are well-established for EEG signals termed as delta, theta, alpha, beta and gamma with each having specified ranges and significance. Fourier transform analysis is done thereby obtaining the five medically ascertained EEG bands – delta, theta, alpha, beta, gamma frequencies. To obtain the amplitude of EEG Signals in these ranges, power spectral density is integrated over the desired frequency range. The time-domain EEG data are converted into the frequency domain using a Fourier Transform (FT), which allows the signals to be broken down into defined frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–100 Hz). Different cerebral activity correlates with each frequency band. Pathological situations like seizures are frequently associated with lower-frequency bands (delta, theta).

The energy distribution across these bands is measured using Power Spectral Density (PSD), which offers crucial indicators for seizure diagnosis. Accurate measurement of signal amplitudes within the intended frequency ranges is made possible by PSD integration across the frequency bands. EEG recordings can show abnormal electrical discharges called epileptiform discharges, which are commonly associated with epilepsy. These discharges can show up as spikes, sudden waves, spike-and-wave complexes, or other irregular patterns in the EEG data. Epileptiform discharges are often described by their shape, duration, amplitude, and geographic distribution. The patients' epileptic episodes are confirmed by the epileptiform discharges detected in the frequency bands.

(c) Data Representation:
(i) 2D Image Representation:

The technique incorporates methods such as the CWT and STFT. These techniques allow for the simultaneous analysis of frequency and time information in EEG signals, which makes it possible to record brief variations in frequency content across time. Through the utilization of STFT or CWT, investigators can derive characteristics that mirror the dynamic evolution of the frequency composition of EEG data, offering significant understanding into the temporal dynamics of brain activity during epileptic episodes.

(ii) Time-Series Representation:

For RNNs or their derivatives, such as LSTM or Gated Recurrent Unit (GRU) networks, the raw or pre-processed EEG epochs can be used as input sequences. RNNs are able to detect epileptic seizures by using the time-series form of EEG data, which allows them to capture temporal dependencies and patterns. By utilizing the advantages of RNN architectures for sequential data modeling, this method makes it possible to identify minute variations and oscillations in EEG signals that may be signs of seizure activity.

(d) Model Selection

CNNs make up the ML model architecture that is used to classify seizures and non-seizures from EEG data. The architecture begins with a 1 dimensional (1D) convolutional layer that uses the ReLU activation function and 32 filters, each with a kernel size of 3. The input EEG signals, which are represented as 1D array with a length of 178, are processed by this layer. A max-pooling layer with a pool size of two is then used to compress the data's spatial dimensions. A max-pooling layer is placed after another convolutional layer with 64 filters and a kernel size of 3. After that, the output of these layers is flattened and fed into layers that are fully connected.

To avoid overfitting, the fully connected layers are composed with a dropout rate of 0.5 in a dropout layer and with 128 neurons in a dense layer and the rectified linear unit (ReLU) activation function. The last layer consists of a single neuron in a dense layer with a sigmoid activation function. It outputs the probability of the input falling into the positive class (seizures) between 0 and 1, which may be interpreted as the chance of having a seizure.

Signals used in EEG analysis consist of several channels that correspond to scalp electrodes and are spatially characterized by nature. CNNs are excellent at identifying spatial relationships in EEG data by applying convolutional filters to extract spatial information.

CNNs identify complicated patterns suggestive of epileptic seizure activity by learning hierarchical representations of spatial patterns. This makes it possible for CNN models to identify and categorize patterns connected to seizures in EEG signals. Utilizing CNNs' spatial capabilities improves seizure detection systems' precision and dependability, enabling prompt epilepsy diagnosis and treatment. As a result, CNN-based methods greatly enhance EEG-based diagnostic methods, providing information about neurological conditions and enhancing patient care in medical settings.

Suitable Hyper parameters to define during training and optimize during testing phase are:

(i) Convolutional Layer Count:
The number of convolutional layers in the CNN architecture is indicated by this parameter. It functions as a key determinant of depth and complexity of the network, which has a direct impact on its ability to extract spatial characteristics and hierarchical patterns from input data.

(ii) Filter Size: The filter size denotes the dimensions of convolutional filters applied to input data. Filters slide across data, multiplying each element. Filter size determines the receptive field, or spatial scope, affecting detail capture. Smaller filters focus on finer details, larger ones on broader features, enabling CNNs to learn hierarchical representations at various scales.

(iii) Pooling Size: Pooling is a down-sampling operation in CNNs that reduces the spatial dimensions of feature maps. The pooling size refers to the dimensions of the pooling windows used for operations like average or maximum pooling. These windows slide over the feature maps, and at each position, they aggregate information based on the specified operation within their dimensions. Adjusting the pooling size affects the amount of information retained in the down-sampled feature maps, influencing the network's ability to capture spatial relationships and reduce computational complexity.

(iv) Activation Function: The activation function, like ReLU or tanh, determines the output of a neuron given its input. It introduces non-linearity, allowing neural networks to learn complex patterns. Applied after each convolutional layer, it helps the network learn and model intricate relationships within the data, enhancing its representation capabilities.

(v) Dropout Rate: The dropout rate is a regularization technique used to prevent overfitting in neural networks. It randomly drops a proportion of neurons during training, forcing the network to learn redundant representations, thus improving generalization and reducing reliance on specific neurons.

(vi) The learning rate: The rate of learning is a critical parameter governing the pace at which a machine learning model adapts its weights during training. It determines the size of the step taken in the direction opposite to the gradient of the loss function. A higher learning rate may lead to faster convergence but risks overshooting the optimal solution, while a lower rate may slow down convergence.

(vii) Batch Size: The batch size refers to the number of data samples processed in a single iteration before updating the model's parameters during training. Adjusting the batch size affects the efficiency of the training process and the memory requirements, with larger batch sizes leading to faster training but requiring more memory.

(viii) Epochs: Throughout the training phase, the network processes the entire dataset a total of epochs. Until the required performance level is attained, all training samples are run through the network during each epoch, and the model's parameters are adjusted depending on the computed error.

(e) Model Training

Several essential steps are incorporated in the ML model's training process. Using the train_test_split function from scikit-learn, the dataset is first divided into training, validation, and test sets. Of the original data, the training set has 60 % of the data, while the test and validation sets each contain 20 % of the data. This guarantees that throughout both training and validation, the model's performance is assessed on untested data.

To make the data compatible with the TensorFlow framework, it is separated and then transformed into TensorFlow tensors. The 'Sequential' application programming interface (API) from TensorFlow.keras is used to design the architecture of the CNN model. Convolutional layers, max-pooling layers for feature extraction, and fully connected layers for classification make up the architecture. One neuron with a sigmoid activation function for binary classification makes up the output layer. The model is then compiled using the Adam optimizer and binary cross-entropy loss function, which are common choices for binary classification tasks. Additionally, accuracy is used as a statistic to track the model's success throughout training.

During the training phase, the \`fit\` approach is used on the model with the training and validation data. The model is trained with a batch size of 32 over 10 epochs. In order to minimize the loss function, the model's weights are iteratively updated during the training process using gradient descent and back propagation. After training, the model's performance is evaluated on the test set using the \`evaluate\` method. This gives the model's ultimate accuracy on hypothetical data. Lastly, performance metrics including precision, recall, F1-score, and the confusion matrix are produced to evaluate the model's classification performance after the predictions are derived using the test data.

The first three metrics collectively offer a comprehensive assessment of the model's predictive accuracy, particularly in imbalanced datasets where the distribution of classes is unequal, and the confusion matrix helps in visualizing the model's classification results.

- Split the segmented EEG data into training, validation, and test sets.
- Train the CNN model with the training dataset. Utilize the validation dataset to alter and adjust hyper parameters and prevent over fitting.
- Use appropriate loss functions such as binary cross-entropy for binary classification (seizure vs. non-seizure) as shown in Eq. (9).

Binary Cross Entropy

$$\frac{1}{N}\sum_{i=1}^{n}(y_i\log(\hat{y}_i)) + (1-y_i)\log(1-\hat{y}_i) \quad (9)$$

Where,

$N \rightarrow$ the sample size

$y_i \rightarrow$ the true label sample i (0 or 1)

$\hat{y}_i \rightarrow$ the predicted probability

It uses negative log likelihood which penalizes the model more heavily if any outlier prediction is made.

As explained in Eqs. (10), (11), (12) and (13) respectively, accuracy, precision, recall and F1 score are the metrics used to evaluate the model's results. These metrics help in identifying how well the model classifies the dataset

$$Accuracy = \frac{Number\ of\ Correct\ Predictions}{Total\ Number\ of\ Predictions} \times\ 100\% \quad (10)$$

Precision can be denoted as

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

where the number of false positives is represented by 'FP' and the number of true positives by 'TP'. The percentage of positive estimates that truly fall into the positive category is known as precision.

Sensitivity, also known as recall, where 'FN' denotes number of false negatives, can be described as

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

The F1 score is calculated using the following formula:

$$F1 = \frac{2\ \times Precision \times Recall}{Precision + Recall} \quad (13)$$

The F1 score is useful for evaluating the overall effectiveness of a classification model, especially in scenarios where both recall and accuracy are important.

(f) XAI Techniques

The SHAP library is used in the XAI integration process within the machine learning model. The SHAP library is used to interpret the ML model's output by connecting the prediction to the input features. It is a powerful technique used to provide insights into the inner workings of machine learning models.

SHAP's model agnostic nature made it a natural choice for this research. Due to its adaptability, the main model doesn't need to be changed to fit the CNN architecture utilized in the study. Both local and global interpretability are offered by SHAP. This facilitates in assigning the significance of every feature in the dataset globally, which aids analysts in comprehending the CNN's overall behaviour. By measuring each feature's contribution to a particular output, it also provides a local explanation of unique predictions, which is crucial for patient-specific diagnosis.

SHAP begins by examining each feature's CNN prediction. A classification probability for seizures or non-seizures is produced by the model. The contribution of each individual EEG datapoint from particular electrodes is then calculated by SHAP by observing how the CNN's output varies based on whether distinct attributes are included or excluded from the input.

The Shapley value, or average marginal contribution of a feature across all potential feature subsets, is determined via SHAP. This guarantees that each feature's impact is taken into account together with various feature combinations, providing a thorough justification of each feature's significance.

The aim is to perform the following tasks in this study:

(i) Model Explanation:

The test data and the learned machine learning model are used to initialize the SHAP explainer. The SHAP values produced by this explainer show how each variable affects the prediction of the model's outcome.

(ii) Visualization of SHAP Values:

To see how input features and model output relate to one another, a variety of charts are used to depict SHAP values. The summary plot, which displays the distribution of SHAP values for each feature over all samples, is one example of this type of visualization. It assists in determining which features most significantly affect the predictions made by the model.

(iii) Feature Dependence Plot:

Another plot generated is the dependence plot, which illustrates how a single feature affects the model's output. This plot facilitates comprehension of the connection between a particular feature and the model's prediction. Analyzing a feature dependence plot allows stakeholders to identify which EEG data points have the most significant influence on the model's predictions. This information is crucial for understanding the underlying patterns captured by the model and can help domain experts validate the model's behavior against domain knowledge.

(iv) Cumulative dependency Plot:

In order to see how features affect the output of the model cumulatively, a cumulative dependence plot is lastly made. This plot, which provides information about the overall contribution of features to the model predictions, displays the cumulative sum of absolute SHAP values for particular characteristics for each sample.

(v) Validity and interpretability of the model:

Incorporating XAI techniques enhances our comprehension of the dataset and its importance within the model's decision-making process. This phase involves scrutinizing the model's consistency, interpretability, conducting error analysis, and refining the model based on the insights gleaned. The main priority is to ensure that the explanations provided are intuitive, comprehensible, and consistent with domain knowledge.

By evaluating the explanations, it is ascertained whether they effectively encapsulate the factors that influence the model's predictions. This comprehensive examination allows us to validate the model's behaviour, identify areas of improvement, and refine the interpretability of the system. Ultimately, this iterative process fosters greater transparency, trust, and confidence in the model's outcomes, promoting its practical applicability and utility in real-world scenarios.

4. Results and discussions

As a part of the study, a Python program was developed to obtain the dataset consisting of EEG recordings from various electrode placements over time duration of 1 second. Each sample comprises 178 data points representing EEG signal values. The dataset includes categories from 1 to 5 denoting different conditions, such as seizure activity, epileptogenic zone records, healthy area records, and recordings with eyes closed or open respectively.

The robustness and generalizability of the suggested model are ensured by the work's extensive training, validation, and testing using a number of publicly accessible EEG datasets. 500–1000 recordings covering a wide range of seizure and non-seizure occurrences are included in each dataset. A comprehensive preprocessing approach was used to maximize signal quality and enable efficient feature extraction. Segmenting continuous EEG recordings into 1-second epochs, using bandpass filtering to reduce noise while maintaining crucial frequency bands linked to seizure activity, and standardizing the signals to guarantee consistent amplitude representation across recordings were important procedures. The model's input consistency is improved by this careful preprocessing approach, which drastically lowers variability.

Google Colab, equipped with an Intel Xeon CPU with two virtual CPUs (vCPUs) and 13GB of RAM, was used for both the model's assessment and training. This allowed for the effective execution of the intricate calculations that are a feature of deep learning frameworks. The CNN model was able to categorize seizure occurrences from EEG signals more effectively due to the integration of several datasets, thorough preprocessing techniques, and intentional data augmentation.

Several data augmentation methods were used to improve model performance even further and reduce overfitting. These methods included using random scaling to boost the dataset's variability, introducing Gaussian noise to imitate various recording settings, and performing temporal shifts to offer multiple viewpoints on the EEG signals. This kind of augmentation not only improves the quality of the training dataset but also strengthens the model's capacity to generalize well to new data while testing.

The EEG signals from different categories were plotted to visualize their patterns and characteristics. The plots showcase distinct variations in signal amplitude and frequency across different conditions, providing insights into the nature of EEG recordings associated with each category as shown in Fig 3. The visualization highlights the differences in EEG patterns between seizure and non-seizure activities, aiding in understanding the characteristics of epileptic signals.

To simplify further analysis, the dataset was relabeled to classify seizure activity as 1 and non-seizure activity as 0. This binary classification enabled the distinction between epileptic and non-epileptic recordings, facilitating further analysis and classification tasks. The total mean and standard deviation values for epileptic and non-epileptic records were computed and were found to be 290.129± 53.563 μV (micro Volts) and 1260.099± 15.561 μV (micro Volts) respectively.

This provided an overview of the central tendency and variability of EEG signals in each category. Descriptive statistics including count mean, standard deviation, and quartiles were calculated separately for epileptic and non-epileptic records. These statistics offered a detailed understanding of the distribution and properties of EEG signals within each class.

In Fig 4, a bar chart was plotted to visualize the count of each category (seizure and non-seizure) in the dataset. This visualization highlighted the distribution of records across different classes and identified any potential class imbalances.

EEG signals for both seizure and non-seizure records were compared through visualizations. Three examples of seizure EEG signals and three examples of non-seizure EEG signals were displayed side by side in Fig 5.

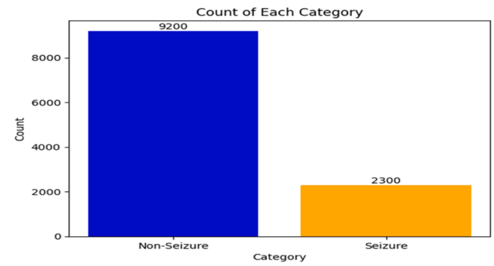

Fig. 4. Classification of Dataset into Seizure and Non-Seizure Classes.

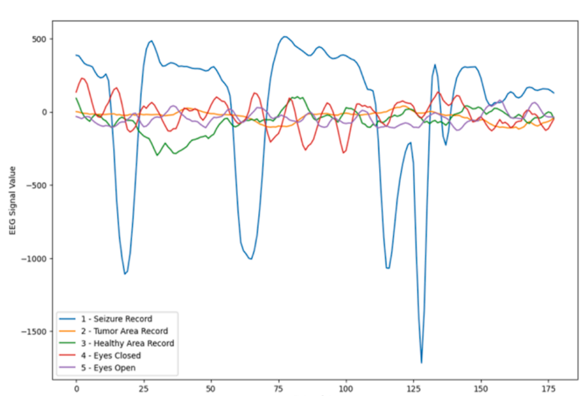

Fig. 3. Visualization of Epileptic Signals in Dataset in Each Category.

Seizure EEG signals are typically characterized by irregular, high-amplitude fluctuations, often with abrupt changes or spikes, while non-seizure EEG signals tend to exhibit more regular patterns with lower amplitude fluctuations and minimal spikes. This comparison facilitated the identification of distinct patterns between the two classes.

In order to obtain exceptional precision in classification for EEG data, the CNN architecture was meticulously adjusted. To maximize important parameters, a grid search strategy was employed in conjunction with hand fine-tuning. The final design included three layers with increasing filters (32, 64, and 128) after the number of convolutional layers were gradually raised. In order to minimize dimensionality without losing important information, max pooling with a size of two was used, while the 3×3 filter size was used since it could catch tiny fluctuations in the EEG signal. Throughout, the Rectified Linear Unit (ReLU) activation function was employed due to its effectiveness in addressing vanishing gradient problems.

The Adam optimizer, which has been demonstrated to provide a suitable compromise between convergence speed and accuracy, was used to provide reliable training at a learning rate of 0.001. Early stopping determined by validation loss was used to avoid overtraining, and dropout layers with a rate of 0.5 were introduced to reduce overfitting. CNN was chosen over RNN because it was more effective at capturing spatial patterns in EEG data, whereas RNNs did not significantly better on this task.

The CNN model attained a final test accuracy of 98.08 %, demonstrating commendable precision on both training and validation sets. The model's training accuracy rose gradually from 72.85 % to 97.98 % across ten epochs, while its validation accuracy hit 97.93 %. These findings highlight the model's accuracy in differentiating between seizure and non-seizure situations.

As shown above in Fig 6, the precision, recall and F1 score are 0.99, 0.91 and 0.95 respectively for classification as seizure, and 0.98, 1.0 and 0.99 respectively for classification as non-seizure. The training and integration results of the CNN model provide significant new insights into the model's interpretability and performance in identifying epilepsy. The accuracy of the CNN model is 98.08 %, which is very satisfactory according to the precision, recall, and F1-score metrics in the classification report.

This study's use of performance metrics—accuracy, precision, recall, specificity, and F1-score—offers a strong foundation for assessing how well the CNN model performs in the identification of epileptic seizures. Although accuracy measures how accurate forecasts are overall, it is insufficient in situations where there is a class disparity. In order to ensure that the model successfully avoids needless interventions in non-seizure occurrences, precision is used to reduce false positive rates. Recall gauges how sensitive the model is to real seizures, reducing the possibility of missed diagnoses—a crucial factor in clinical practice. In situations when there is a class imbalance, the F1-score provides a comprehensive metric that balances the trade-offs between precision and recall by offering a harmonic mean of both metrics. Finally, specificity measures how well the model can detect non-seizure cases, which

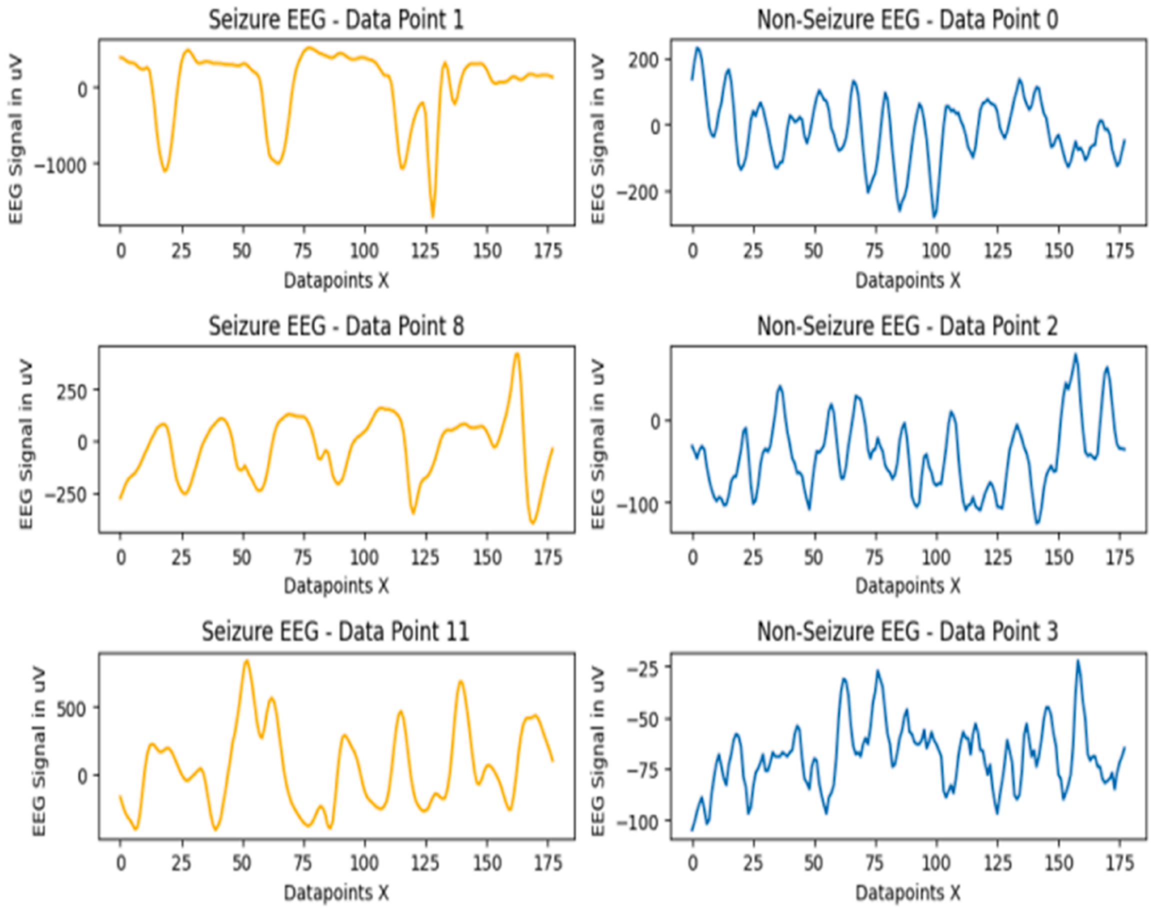

Fig. 5. Plots comparing Non-Seizure EEG to Seizure EEG for particular datapoints in the dataset.

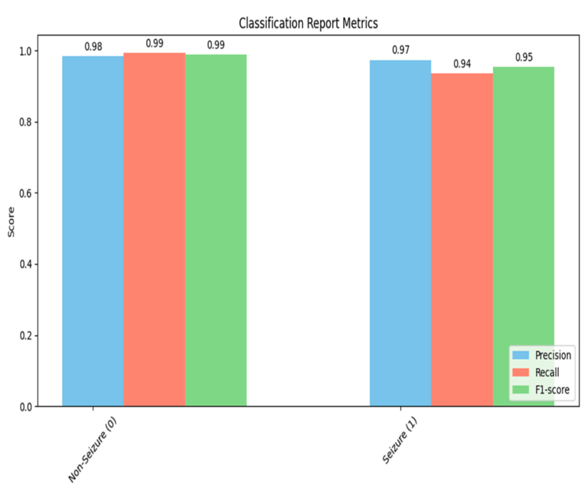

Fig. 6. Classification report metrics of CNN model.

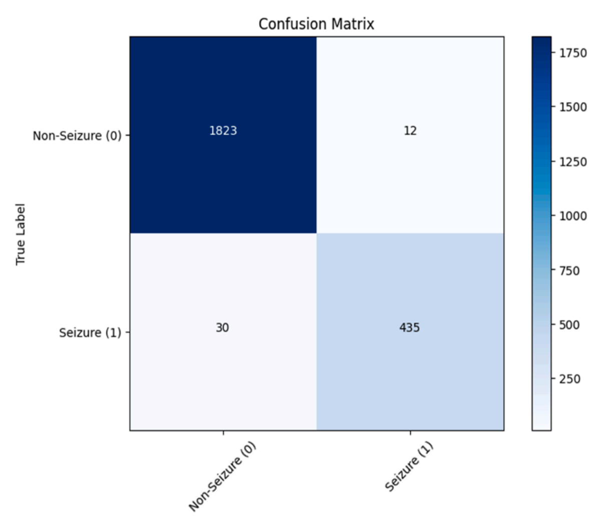

Fig. 7. Confusion matrix for the CNN model results.

is essential for avoiding over-diagnosis. When combined, these indicators improve the model's interpretability and dependability and guarantee that it satisfies the exacting requirements of clinical seizure detection applications.

The model's performance in distinguishing between seizure and non-seizure situations is displayed in the confusion matrix as shown in Fig 7. The model shows a high degree of accuracy in recognizing seizure and non-seizure cases, with 1832 true negative predictions and 424 true positive predictions, respectively. The total performance is still quite good, despite the few false predictions due to practical issues.

The metrics highlight the model's dependability in clinical situations. Clinicians would be able to gain confidence from the model's result as it the predictions made by it correlate with the actual ground truths in 98 % of the cases, thereby attesting to the model's merits.

When assessing the effectiveness of classification models, especially when it comes to seizure detection using electroencephalogram (EEG) signals, the AUC-ROC is a crucial statistic. As shown in Fig 8, a thorough

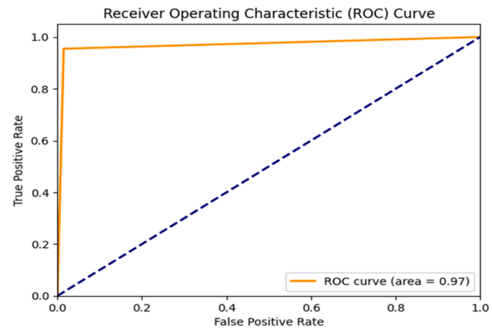

Fig. 8. AUC-ROC Curve for Seizure Detection Model.

picture of the model's performance across all classification thresholds is provided by the ROC curve, which illustrates the trade-off between sensitivity (true positive rate) and specificity (false positive rate) at different threshold values. The high degree of separability between the positive (seizure) and negative (non-seizure) classes is shown by an AUC value of 0.97, highlighting the model's effectiveness in precisely detecting seizure episodes while reducing false positives.

The convolutional neural network (CNN) utilized in this study for seizure detection is made easier to understand by the AUC-ROC value, which also acts as a reliable indicator of model performance. By confirming the efficacy of the used features and the overall design, a high AUC-ROC score validates the clinical usability of the model. Additionally, the interpretability provided by explainable AI techniques like SHAP combined with the AUC-ROC analysis guarantees that physicians can trust and comprehend the model's predictions, which eventually improves patient outcomes in the treatment of epilepsy.

A rough structure of electrode placements during EEG data collection is shown in Fig 9 which gives an idea on how brain in segregated into frontal, central, parietal, occipital regions in order to detect seizures.

A logistic regression model is used as a baseline model to compare the suggested CNN model's performance against it. Confusion matrices, classification metric bar graphs, and ROC curves are used to visually compare the CNN model with the baseline model, offering a thorough assessment of the models' performance.

(i) Confusion Matrices: The CNN model's increased classification accuracy is demonstrated by the confusion matrix findings as shown in Fig 10. In addition to a notable decrease in false negatives (30 vs. 407), it demonstrates greater true positives for both "Non-Seizure" (1823 vs. 1833) and "Seizure" (435 vs. 58) instances. This suggests that the CNN model has a noticeably higher recall than the baseline, indicating that it can identify actual seizure occurrences with greater accuracy.
(ii) Classification Metrics Comparison: The bar plot, Fig 11, displays a clear difference between the two models when comparing the seizure class's precision, recall, and F1-score. The accuracy of both models is comparable (~0.97), indicating that they are able to correctly categorize seizure cases when they are anticipated to be positive. However, the CNN model's recall (0.94) far outperforms the baseline model's (0.12), highlighting the CNN's capacity to detect a substantially greater percentage of real seizure occurrences. The CNN model's F1-score is significantly higher as a result (0.95 vs. 0.22), indicating a balanced performance in terms of recall and precision. These findings suggest that the CNN model is significantly better at properly identifying the majority of real seizure occurrences.
(iii) ROC Curve Analysis: The ROC Curve in Fig 12 highlights the CNN model's discriminative ability. A greater true positive rate for each false positive rate is indicated by the CNN model's ROC curve staying around the top-left corner, in contrast to the baseline model, whose curve deviates greatly from the optimal route. The CNN model has a significantly bigger area under the ROC curve (AUC), indicating that it can more effectively differentiate between seizure and non-seizure cases over a range of decision criteria. The CNN model's increased AUC demonstrates its better generalization and robustness compared to the baseline model, which increases its dependability for clinical use.

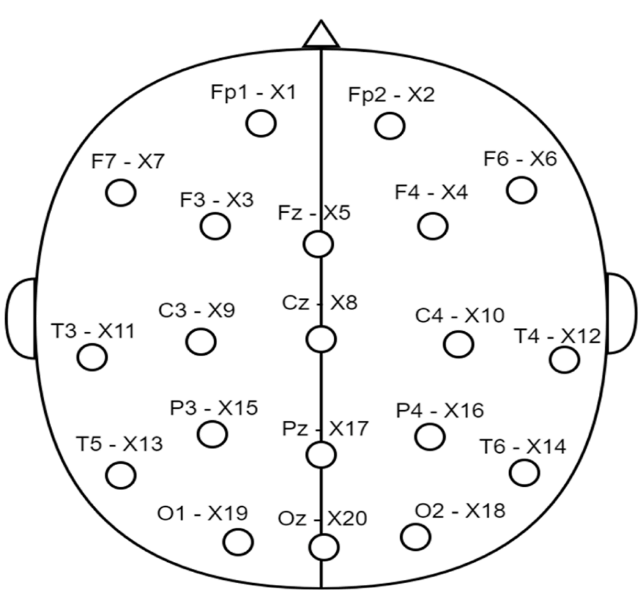

Fig. 9. Sample Diagram of Electrode Placements.

From these visual comparisons, it is evident that the CNN model significantly outperforms the baseline model in terms of classification accuracy, sensitivity, and overall discriminative performance, all of which are crucial for the real-world implementation of seizure detection systems.

In summary, the proposed CNN-based model demonstrates excellent performance in epileptic seizure detection with high accuracy rates and interpretability through XAI techniques. By employing SHAP values, the study successfully elucidates the significant features that contribute to seizure prediction, offering clinicians valuable insights. The model's capability to integrate EEG data from different sources, combined with real-time potential, enhances its application in clinical workflows. Furthermore, the incorporation of hybrid models and attention mechanisms indicates a promising direction for future research aimed at improving the robustness and real-time capabilities of seizure detection systems. The findings suggest that deep learning architectures, when carefully optimized, can provide reliable support for epilepsy management. Future efforts should aim to strike a balance between maintaining

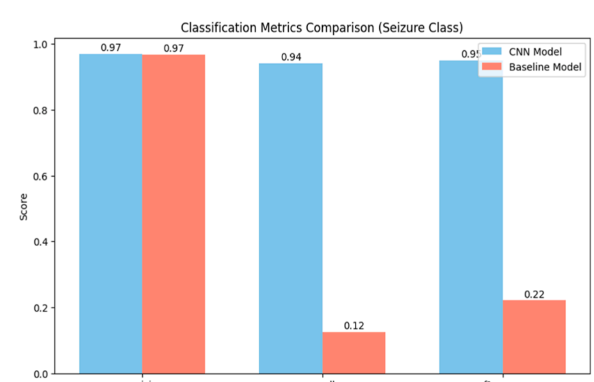

Fig. 10. Comparison of Classification Metrics of CNN and Baseline Model.

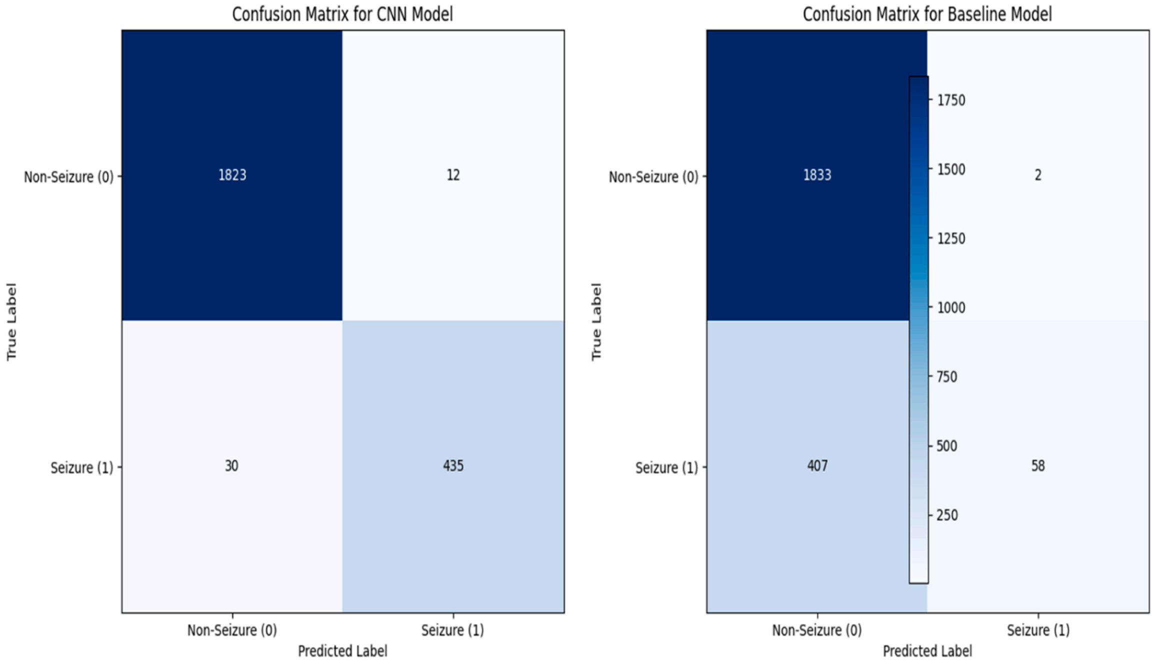

Fig. 11. Comparison of Confusion Matrix of CNN and Baseline Model.

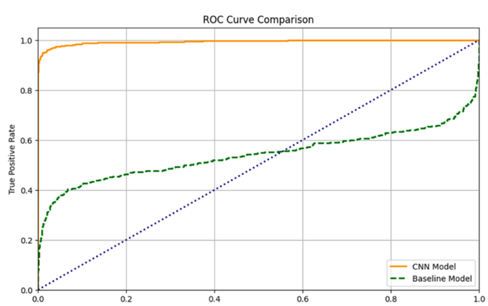

Fig. 12. Comparison of AUC-ROC Curve of CNN and Baseline Model.

high accuracy while reducing computational complexity, to ensure practical, scalable applications in healthcare settings.

The SHAP summary plot, shown in Fig 13, provides information on how specific characteristics (electrode datapoints) affect the output of the model. To improve interpretability, SHAP was used to incorporate Explainable AI (XAI), which offered localized insights into how particular EEG variables affected predictions. By recognizing the significance of specific variables, which are necessary for precise seizure identification, this helps medical practitioners to trust the model's decision-making process in clinical situations. Clinicians are able to recognize important EEG signals that are involved in seizure prediction by displaying feature significance ratings. In order to facilitate the understanding of the model's conclusions and focus future research into pertinent physiological processes, the violin plot distribution illustrates the strength and direction of each data point's effect.

The data points that are skewed towards the right of the mid-zero line, such as data point 178 indicate positive influence to the model's predictions. They have a high impact on the outcomes while on the other hand, the data points biased towards the negative side of the midline have relatively lesser contribution towards the results of the model and comparatively, are not very significant. It can be seen that the datapoint X178 has the highest impact on the model's output, followed by X89.

A dependence plot, on the other hand, focuses on the relationship between a specific feature (say Datapoint 1, or X1) and its corresponding SHAP values. This plot illustrates how changes in the feature represented by Datapoint 1 influence the model's output, as indicated by the SHAP values. The plot visually represents the impact of the feature on the model's predictions, helping to interpret the model's behavior and understand the importance of the feature in the decision-making

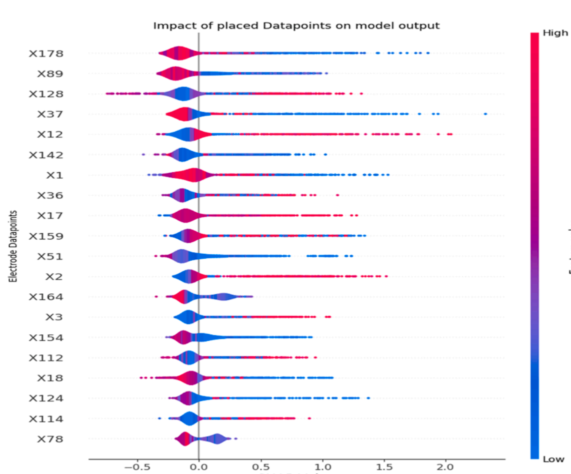

Fig. 13. SHAP Summary Plot of Datapoints In Model Result.

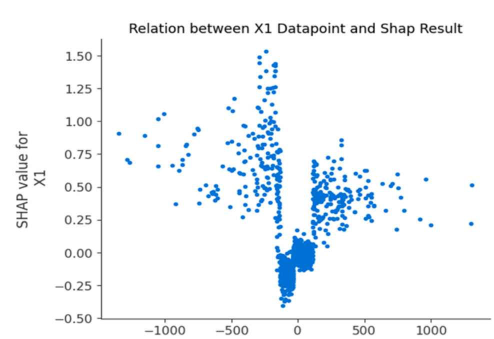

Fig. 14. Dependency plot of X1 with regards to SHAP Values.

process. Fig 14 illustrates the influence of fluctuations in X1 on the SHAP value output. It can be seen how deviation of X1 value from 0 increases the shap value, since it suggests possible seizure activity.

This graph provides insights into the physiological importance of EEG data by illustrating how variations in one particular feature value impact the model's output probability. By identifying important patterns or trends linked to seizure prediction, therapists can utilize this information to observe how certain electrode placements in specific cranial areas can contribute more to the detection of seizures. This paves way to have more informed targeted treatments and interventions.

A cumulative dependence plot is used to describe the total significance of a datapoint on the model output across the dataset as the sample size increases. Fig 15 shows the cumulative dependence plot of the most and least important feature. Here the datapoint 70 shows minimum contribution to the model's prediction and datapoint 177 shows maximum contribution as evident by the nature of their plots.

A succinct yet informative indicator of how much an electrode data point—a single feature—influences the CNN model's predictions is provided by the mean SHAP values. Regions of increased relevance in seizure detection are indicated by features with higher mean SHAP values, for instance, Electrode Datapoints 177 and 88. These values offer practical information to physicians regarding the particular EEG signals that have the most influence on the model's decision-making process. Physicians may optimize the diagnostic workflow and improve the successful outcome of epilepsy control procedures by giving priority to features with increased mean SHAP values. This allows physicians to concentrate their attention on crucial areas of EEG data. Additionally, mean SHAP values' interpretability encourages confidence in the model's predictions, enabling physicians to make defensible judgments based on clinically meaningful EEG data. Enhancing both diagnostic precision and therapeutic options, this interpretative framework eventually leads to better clinical results in handling cases of epilepsy.

In order to understand the predictive performance of the CNN model with reference to epileptic seizure detection, the cumulative dependency plot is a significant tool that can be leveraged. This plot makes it clear on how different characteristics affect the model's decision-making process as a whole. Interestingly, several features—like Electrode Datapoint 177—show strong contributions, suggesting that they are critical to the model's estimates. This is seen from the steep incline and steady growth of Data point 177 plot while the number of samples increases consistently. On the other hand, the cumulative SHAP values of other variables show more gradual rises, indicating their minor significance in seizure detection. Clinicians can implement targeted interpretation and intervention techniques by identifying these patterns, which provide a detailed knowledge of the model's dependence on certain EEG signals.

Furthermore, the model's stability and generalizability are critically assessed using the cumulative dependency plot. The ability to capture prominent patterns suggestive of seizure activity is demonstrated by

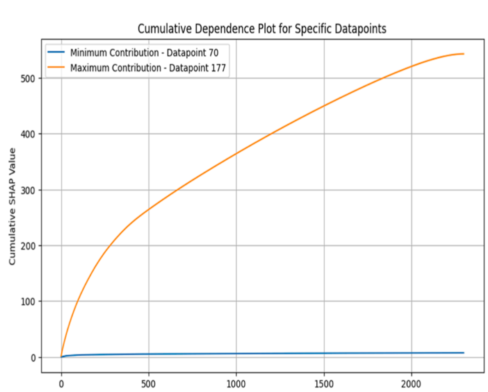

Fig. 15. Cumulative dependence plot for maximum and minimum datapoints in the model results.

persistent trends in aggregate SHAP values across samples. This consistency makes the model's predicted performance more credible to doctors, making it easier for the model to be integrated smoothly into clinical workflows. Thus, by using the CNN model's trustworthy forecasts, medical professionals may speed up diagnostic and treatment choices, which will eventually enhance patient results and the quality of healthcare.

The performance and comprehensibility of the CNN-based epileptic seizure detection model have been thoroughly examined providing insight into the model's possible uses in practical clinical settings. The study highlights the resiliency and dependability of the model by carefully examining classification reports, confusion matrices, mean SHAP values, cumulative dependency graphs, and SHAP summary plots. The results emphasize how important it is to use cutting-edge machine learning methods and XAI approaches to improve diagnostic precision and expedite clinical decision-making in the treatment of epilepsy. To enable the model's widespread use in clinical practice, future research efforts should concentrate on improving the model's performance and building confidence among healthcare professionals.

Computational overhead is one of the challenges in the suggested CNN-based epileptic seizure detection methodology. Convolutional layers and SHAP values are two methods that might be resource-intensive when used to extract significant features from EEG data, especially for real-time applications. Longer training times and a large computational load are caused by the CNN architecture's many parameters, which may not be practical in settings with constrained hardware.

Furthermore, using ensemble models or hybrid algorithms that combine CNNs with attention mechanisms or RNNs clearly shows the trade-off between accuracy and computing economy. These sophisticated models can improve prediction accuracy, especially in difficult situations like complicated seizure types or pre-seizure states. They do, however, come at the expense of extra layers and parameters, which raises the processing requirements even further. When developing systems for therapeutic usage, striking a balance between these trade-offs becomes crucial.

One of the many obstacles to the clinical application of AI-driven seizure detection models is adherence to strict regulatory requirements, such as FDA and CE-mark approvals for medical equipment. To maintain regulatory approval and clinician trust, the interpretability of the model—which SHAP facilitates—must adhere to transparency standards. Furthermore, processing real-time EEG data presents substantial challenges due to its high computational complexity in contexts with limited resources. Deep CNN architectures' computational overhead may have an impact on real-time decision-making, necessitating improvement before implementing the model in clinical processes.

EEG recordings raise data privacy issues that need to be carefully handled under regulations like GDPR and HIPAA, especially when it comes to patient-specific data. In clinical contexts, robust model validation across a range of patient populations, guaranteeing generalizability and reducing bias, continues to be a difficulty.

5. Conclusion

This study presents a CNN-based epileptic seizure detection model, achieving a test accuracy of 98.08 %, with precision, recall, and F1-scores of 0.99, 0.91, and 0.95, respectively. The model's robustness in differentiating seizure from non-seizure EEG signals is confirmed by the confusion matrix, which shows 1832 true negatives and 424 genuine positives. The most important feature, according to SHAP analysis, was Electrode Datapoint 177, which offered crucial information about the interpretability of the model. Clinicians can have a better understanding of the model's decision-making process because to the SHAP-based feature attribution, which increases the model's usefulness in clinical settings.

The model's efficacy across a variety of EEG datasets is further supported by its high specificity and AUC-ROC. However, real-time applications face difficulties due to the computational complexity, which is fueled by its deep architecture and wide parameter space. Predictive accuracy and computational efficiency have a significant trade-off, especially for deployments with limited resources. Future research should investigate ensemble approaches or hybrid models to maximize performance with little computing overhead.

The model's interpretability is improved by incorporating explainable AI (XAI) approaches like SHAP, which provide lucid insights into the contributions of EEG signals. Because it guarantees that the model's predictions are in line with medical knowledge, this transparency is essential for clinical integration. Despite the model performing well in seizure detection, its scalability still depends on maximizing computing efficiency. Improving computing efficiency for real-time applications should be the main goal of future research.

Author contributions

All the authors have contributed to this article equally.

Data availability

The data used to support the findings of this study are openly available in the paper [31] with DOI: https://doi.org/10.1103/physreve.64.061907.

CRediT authorship contribution statement

Tamilarasi Kathirvel Murugan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anush Kameswaran:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Tamilarasi Kathirvel Murugan reports article publishing charges was provided by Vellore Institute of Technology - Chennai Campus. Tamilarasi Kathirvel Murugan reports a relationship with Vellore Institute of Technology - Chennai Campus that includes: employment. Tamilarasi Kathirvel Murugan has patent NA pending to NA. Kindly consider for publication If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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